BACHELOR THESIS PROJECT II

Solar-Powered Night-Time Cooling Framework

A Software-Driven Approach to Time-Shifting Renewable Surplus via Solar Energy Storage

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Executive Summary

Systems Overview

The Bottleneck

- 5.5 MWp solar infrastructure generates 6 GWh/year (peaking midday).
- Proposed cooling for 8,173 residential rooms peaks at midnight.
- Unmitigated load threatens massive state grid stress.

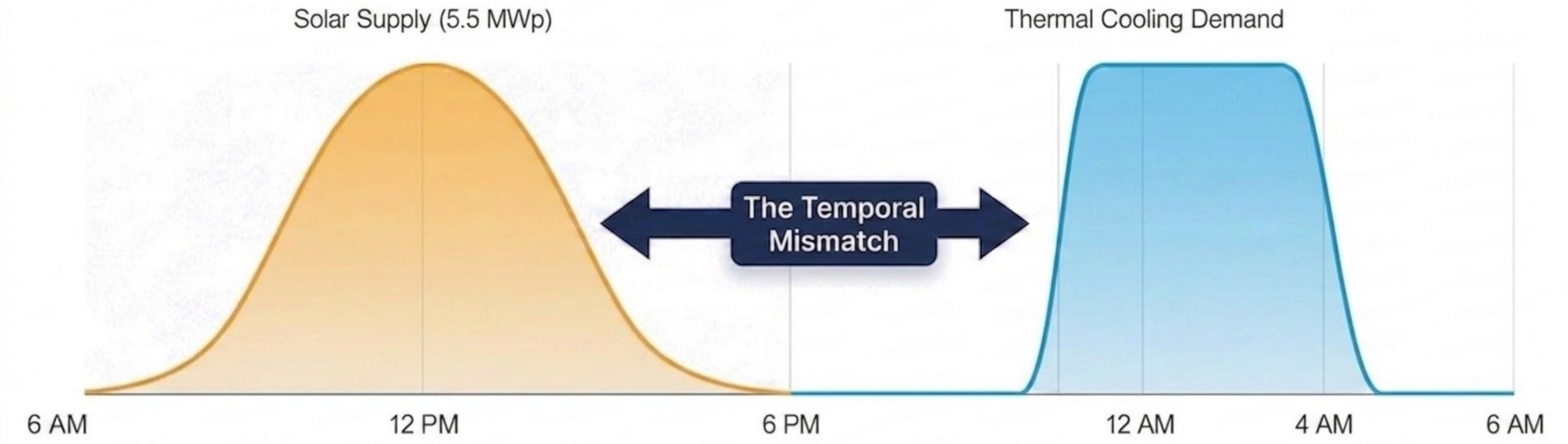
The DSS Intervention

- A Python/Streamlit Decision Support System (DSS).
- Utilizes chilled water Thermal Energy Storage (TES).
- Mathematically time-shifts the daytime Safe Harvest solar surplus.

The Validated Impact

- Achieves equitable thermal comfort across 21 Halls of Residence.
- Utilizes only 53.7% of the daily solar surplus.
- Optimized 60/40 solar-to-grid power blending algorithm.

The Core Challenge: Temporal Misalignment



Daytime Inefficiency

5.5 MWp solar generation is highly underutilized during weekends and vacation periods, feeding back into the grid at lower efficiency.

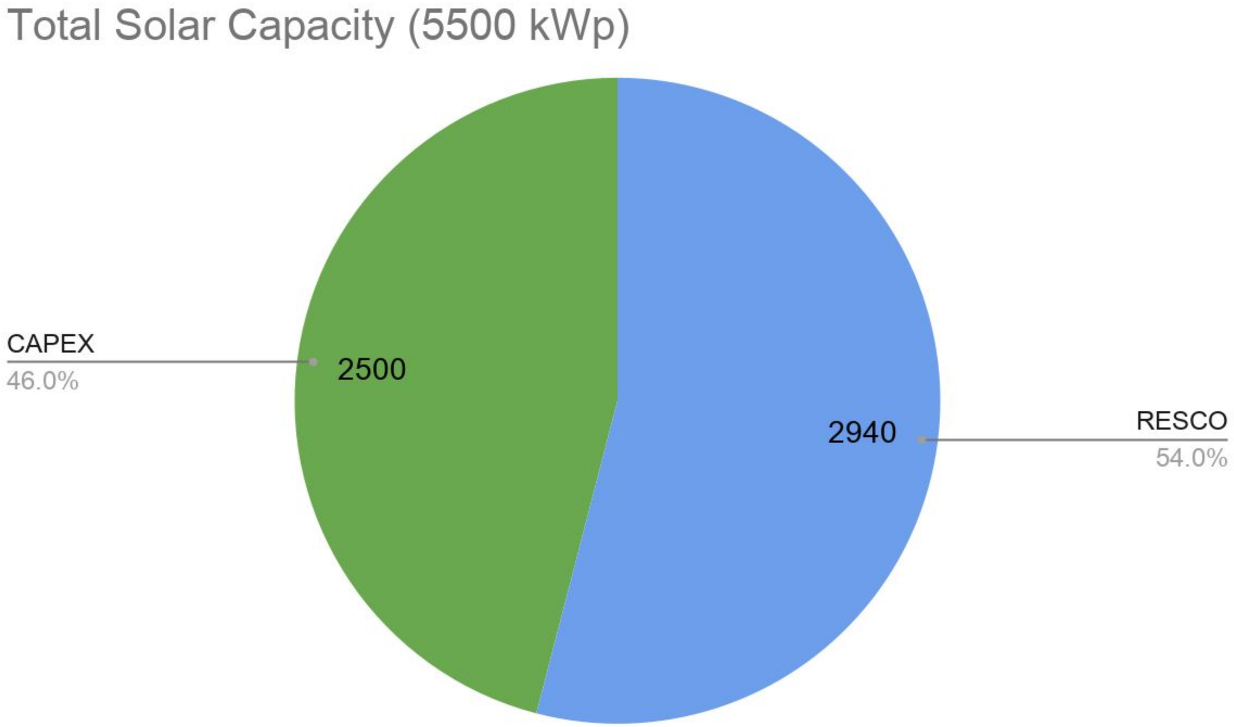
Nighttime Grid Stress

Scaling centralized AC operation to all 21 halls between 10 PM and 4 AM creates an unsustainable instantaneous electrical surge.

The Objective

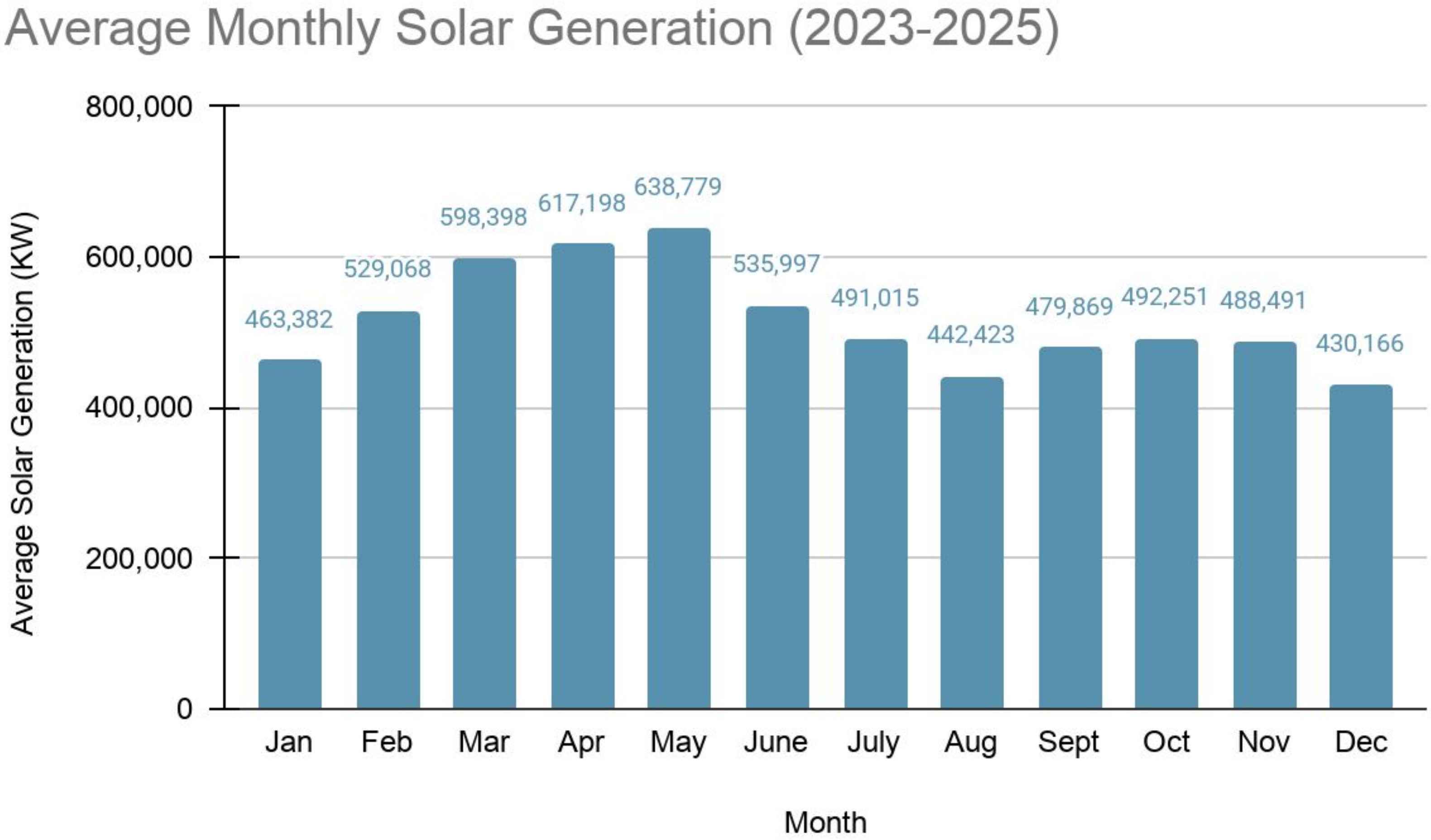
Bridge the gap. Time-shift the Safe Harvest limit into night-time cooling without cannibalizing daytime loads.

Solar Resource Assessment (2023-2025)

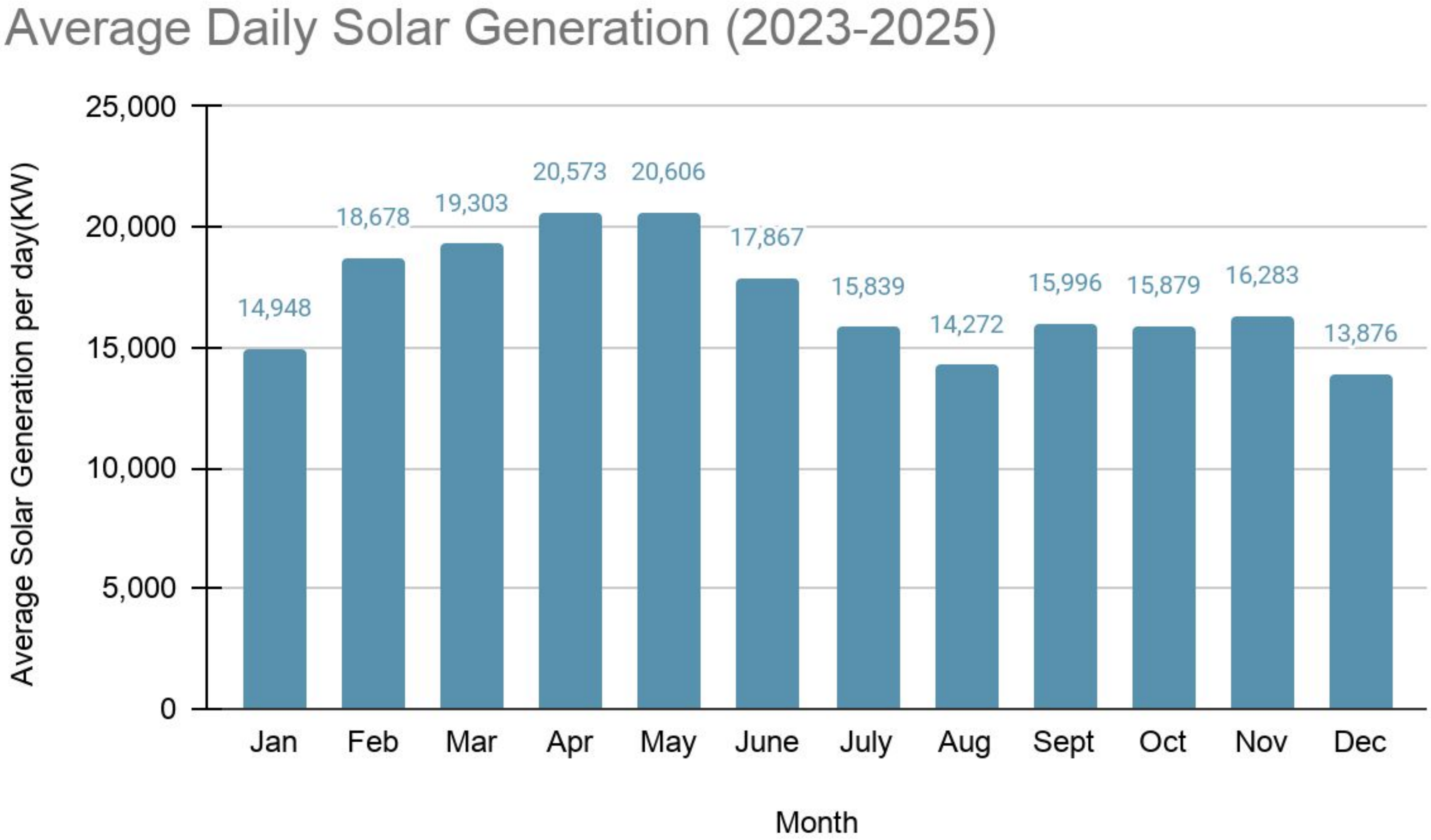


Infrastructure Bifurcation

Solar Resource Assessment (2023-2025)



Solar Resource Assessment (2023-2025)



Annual Operational Phases & Constraints

May-Jul

The Surplus Phase

Max generation (20.6k kWh/day) but minimum load due to summer vacation. High grid export.

Aug-Oct

Critical Design Window

The Worst-Case Scenario. High humidity + full occupancy + Monsoon Dip in solar yield (14.2k kWh/day). System must survive this envelope.

Nov-Feb

The Dormant Phase

No thermal demand. Solar completely supports institutional grid.

Mar-Apr

The Transition Phase

Ideal balance of high solar yield (20k kWh/day) and high occupancy.

MATHEMATICAL FRAMEWORK FOR POWER DEMAND

CORE VARIABLES

Total Rooms (N_{rooms}): Building sum across selected halls.

Cooling Intensity (TR_{room}): Thermal load required per room. (Adjustable: 0.1 to 1.0 TR).

Efficiency (CoP): Coefficient of Performance of chiller plants. (Adjustable: 2.0 to 5.0).

CALCULATION METHODOLOGY

1. Total Cooling Capacity (TR_{total}):

$$TR_{total} = N_{rooms} \times TR_{room}$$

2. Instantaneous Electrical Power Demand (P_{elec}):

$$P_{elec}(kW) = \frac{TR_{total} \times 3.517}{CoP}$$

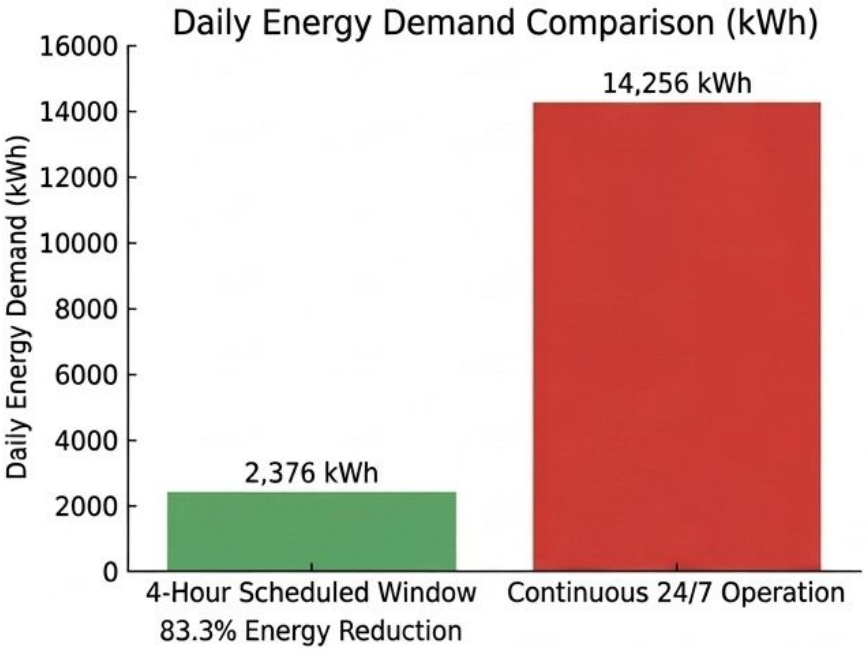
(Conversion: 1 $TR \approx 3.517$ $kW_{thermal}$)

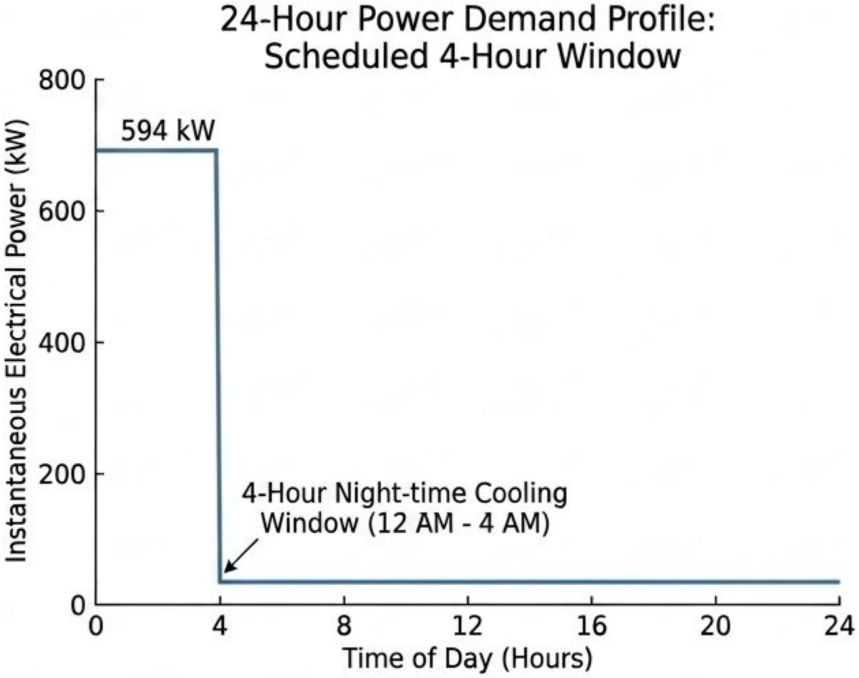
KEY INSIGHTS

- Transition from thermal load to electrical consumption is governed by these room-based variables.
- Framework is calibrated for hardware efficiencies before scaling to the proposed rollout.

Efficiency through Scheduling: 24/7 vs. 4-Hour Operation

Validation of the Time-Shifting Methodology for 900 Baseline Rooms (SBP-1, SBP-2)





This 4-hour scheduled operation is the fundamental driver for framework viability across the entire 21-hall campus proposal.

Storage Capacity Mapping and Variable Scaling

Fundamental Storage Relationship: Sizing for Stored Energy (S_{design})

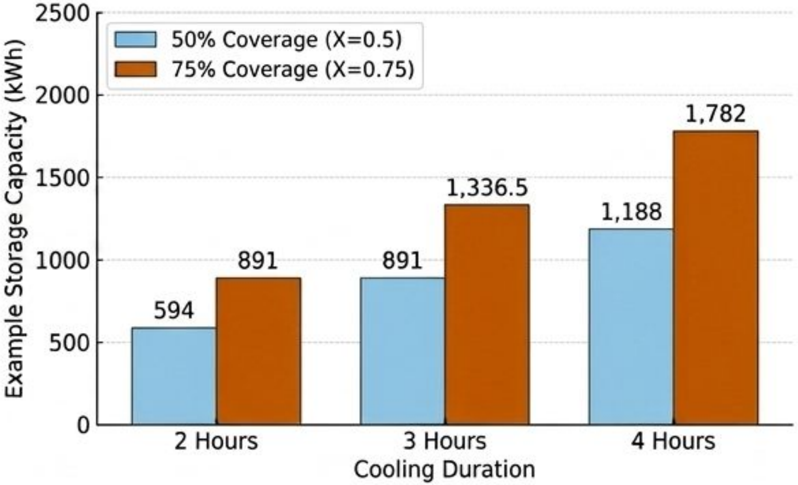
Key equation derived from section 4.2.1 without cites:

$$S_{design}(kWh) = X\% \times (N_{rooms} \times P_{elec\_room} \times t)$$

Where fort are user-adjustable input (no icons):

- **X%**: Load Coverage Factor (Slider, variable, 0 to 100%).
- **N_rooms**: Total Rooms (Adjustable, e.g., 900 baseline).
- **P_elec_room**: Electrical Power per room (kW, Slider, derived from variable TR and CoP).
- **t**: Cooling Duration (hours, Adjustable, 0-24hrs).

Example Case: Illustrative Storage Sizing (900 Baseline Rooms, P_{total} = 594 kW)



Visualization from Figure 4.1 / Table 4.1

Evaluation of Storage Strategies: BESS vs. TES

BESS (Battery Energy Storage Systems): High efficiency (85-90%), but constrained by high CAPEX and limited cycle life.

TES (Thermal Energy Storage) - Proposed Solution: Lower cost per kWh, highly compatible with existing centralized chiller plumbing.

TES Sizing Calculations

$$V(m^3) = \frac{S_{design} \times 3600}{\rho \times c_p \times \Delta T}$$

$\rho = 1000$ kg/m^3
$c_p = 4.186$ kJ/kg.K
$\Delta T = 5$ K

For $S_{design} = 9{,}138$ kWh: $V = 1{,}572$ m^3

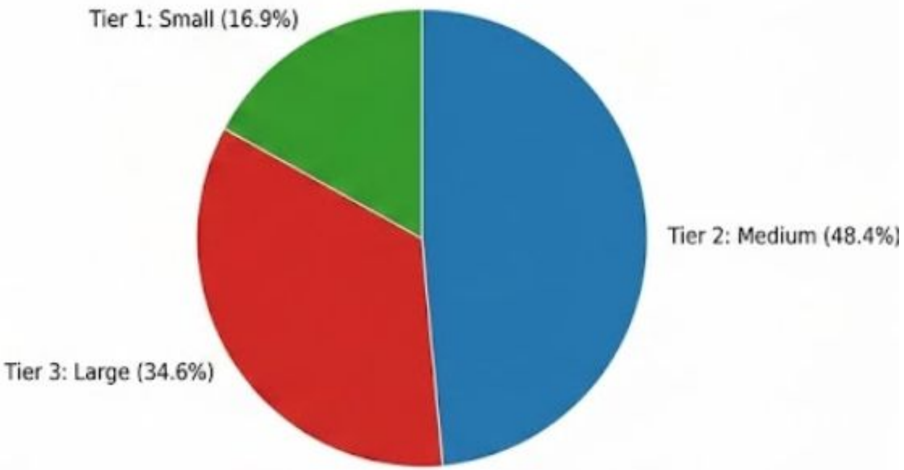
Equivalent to a tank approx 20m diameter and 5m height.

Waterfall Tiered Scheduling Strategy

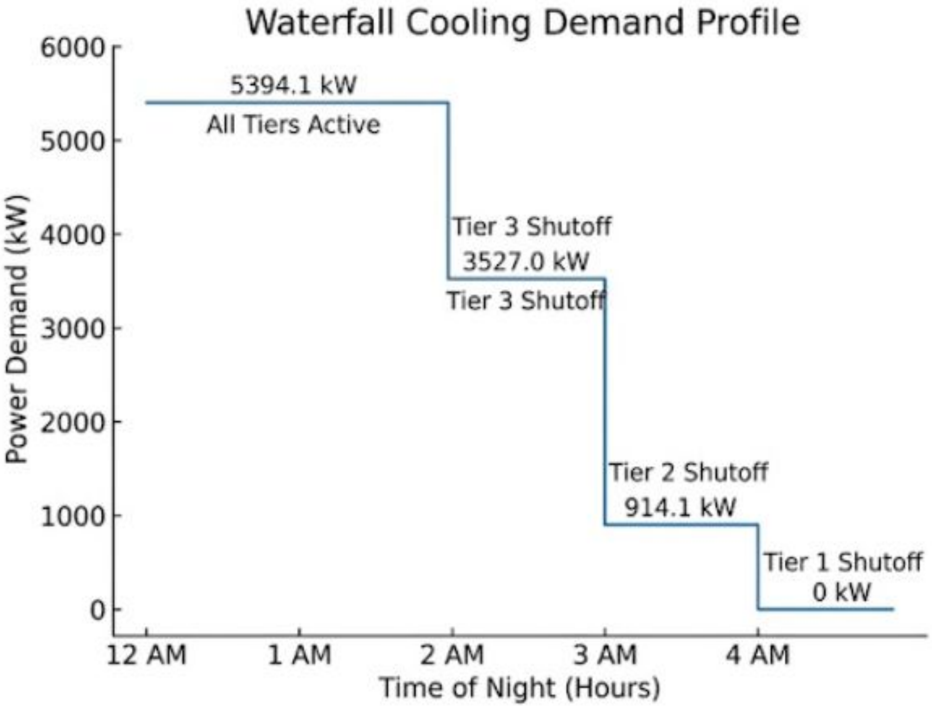
Table 5.1: Tiered Campus Categorization

Tier	Room Count	Duration	Power
Tier 1 (Small)	< 270	4 Hours	914.1 kW
Tier 2 (Medium)	270-600	3 Hours	2,612.9 kW
Tier 3 (Large)	> 600	2 Hours	1,867.1 kW

Distribution of Instantaneous Power Demand
(Total: 5,394.1 kW)



Sequential Tier Shutoff Profile



Total Energy & Power Required

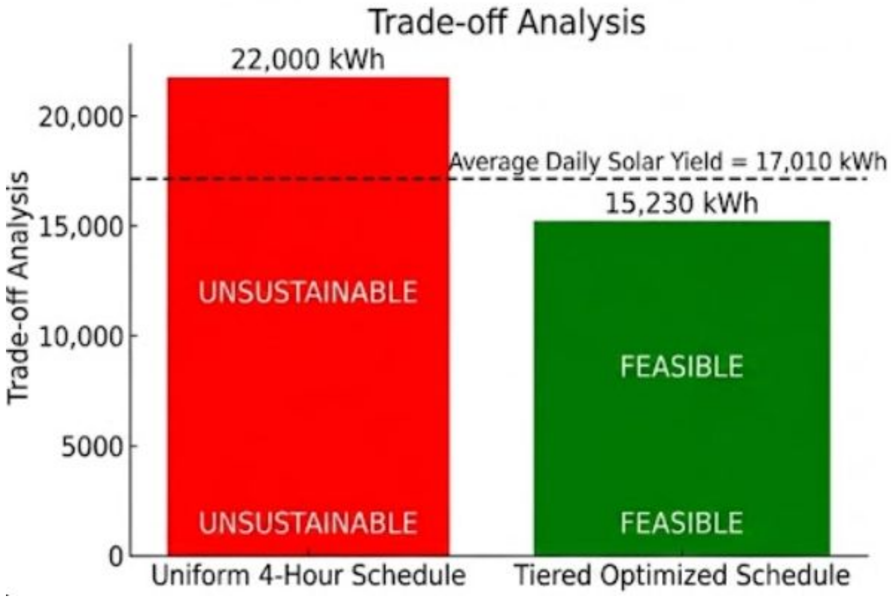
Peak Power = 5,394.1 kW

Formula:
E_total = (P_Tier1 x 4h) + (P_Tier2 x 3h) + (P_Tier3 x 2h)

Calculation:
E_total = (914.1 x 4) + (2612.9 x 3) + (1867.1 x 2)

Final Result:
E_total = 15,229.5 kWh

Final Feasibility & Solar Utilization



Detailed Calculations:

1. Design Storage Capacity (S_{design}) Calculation

$$S_{design} = \text{Optimal Scaling Factor} \times E_{total}\ \text{(Tiered)}$$
$$S_{design} = 0.60 \times 15{,}230\ \text{kWh}$$
$$S_{design} = 9{,}138\ \text{kWh}$$

2. Solar Utilization (%) Calculation

$$\text{Utilization (\%)} = (S_{design} / \text{Daily Solar Yield}) \times 100$$
$$\text{Utilization (\%)} = (9{,}138\ \text{kWh} / 17{,}010\ \text{kWh}) \times 100$$
$$\text{Utilization (\%)} \approx 53.7\%$$

The Software Implementation: Streamlit DSS

Objective: Transition from theoretical data to a live Decision Support System (DSS) for campus administrators.

Module 1: Dynamic Config

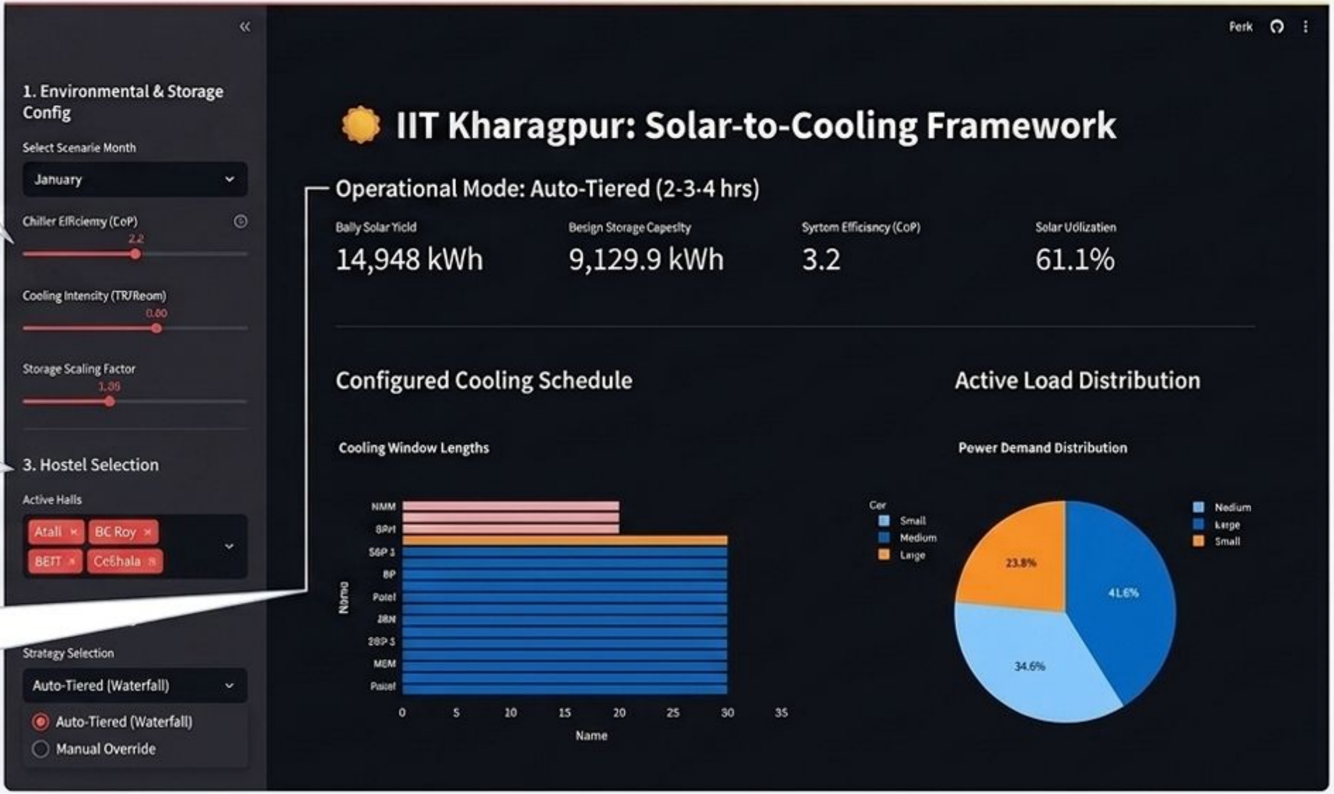
Variable sliders for CoP (chiller efficiency) and TR (cooling intensity) to reflect changing hardware conditions.

Module 2: Hostel Selection

Multi-select tool for targeted, selective energy dispatch (e.g., bypassing halls undergoing maintenance).

Module 3: Strategy Override

Toggle seamlessly between the Auto-Tiered Waterfall logic and Manual Overrides (0-24 hrs) for granular control.



CONCLUSION

The Problem: Temporal Mismatch & Grid Stress

- A fundamental misalignment exists between peak daytime solar generation and high-intensity midnight residential cooling demand.
- Scaling standard centralized cooling to all 21 halls would create massive, unsustainable instantaneous loads on the state power grid.

The Solution: Dynamic Time-Shifting Framework

- **Thermal Energy Storage (TES)**: A physically scalable pathway to store daytime solar surplus as chilled water for night-time circulation.
- **Variable Parameter Modeling**: Moving away from static models to allow real-time calibration of hardware efficiency and cooling intensity.

Final Outcomes & Impact

- **Campus-Wide Equity**: Validates the feasibility of providing 2-4 hours of scheduled nocturnal cooling for all 8,173 rooms.
- **Sustainable Optimization:** Achieves thermal comfort utilizing only ~53.7% of the daily solar surplus, safeguarding daytime institutional loads.
- **Operational Control**: Transitions theoretical research into a live Decision Support System (DSS) for transparent, real-time energy dispatch.

Thank You